

General Waveguide Bend Design Based on Cubic Spline Interpolation and Inverse Design

Enge Zhang , Graduate Student Member, IEEE, Shanglin Yang , and Lei Zhang , Member, IEEE

Abstract—Silicon waveguide bends are key components that affect the packing density and scalability of photonic integrated circuits. Traditional bend design methods rely on well-established functions such as Euler spirals or Bezier curves, which have limited parameter spaces, limiting the discovery of compact bends with higher performance. Here, we propose a general approach to waveguide bend design. For a given radius, our method uses cubic spline interpolation with an inverse design algorithm to obtain an optimal bend geometry to support a given number of modes. We refine the particle swarm optimization algorithm to reduce the need for parameter tuning and increase its ability to avoid local optima. As a result of these modifications, the design process becomes less complicated and more efficient. To demonstrate the effectiveness of the proposed method, we design and fabricate three waveguide bends supporting one, two, and three modes with radii of 2.5 μm , 4 μm , and 5 μm , respectively. The measured (simulated) insertion losses of the three bends are less than 0.012 (0.009) dB, 0.04 (0.028) dB and 0.059 (0.048) dB per 9°, respectively. The measured (simulated) crosstalk between the modes of the latter two bends is less than −23.6 (−36.1) dB and −19.7 (−26.5) dB per 9°, respectively. The proposed method provides an effective tool for designing compact multimode bends for future photonic integrated circuits.

Index Terms—Compact bend, cubic spline interpolation, inverse design.

I. INTRODUCTION

LARGE-SCALE photonic integrated circuits (PICs) have attracted increasing attention and applications in various fields such as high-performance computing systems [1], high-speed communications [2], quantum technology [3] and biosensing [4]. To achieve a more compact layout and reduce cost, high-density integration is required for PICs. Waveguide

bending, a key component of interconnect wiring in PICs, significantly impacts the integration density and signal integrity of the system. A smaller bend radius enables higher layout density, allowing for more devices to fit in a limited space. Fig. 1 presents a schematic of an on-chip tunable optical delay line. To achieve a desired amount of delay time while minimizing the system footprint, a large number of waveguide bends are required to achieve a compact layout [5], [6], [7], [8], [9].

Simultaneously, mode division multiplexing (MDM) technology is an effective method for increasing the capacity of optical interconnect links [6]. MDM technology uses multimode channels in multimode waveguides to significantly enhance the transmission capacity of the system. In recent years, various multimode silicon photonic devices have been developed and applied to MDM systems. These include multichannel mode multiplexers [10], [11], [12], multimode waveguide crossings [13], [14], [15], multimode power splitters [16], [17], and multimode chip-fibre couplers [18], [19]. These devices are crucial to MDM systems and enable the realization of efficient and reliable optical interconnections.

Waveguide bend is a fundamental on-chip interconnect element in PICs and MDM systems. It directly affects the performance, size, and density of the overall photonic device. The performance and radius of waveguide bends are becoming increasingly important due to growing research and application requirements [6], [20]. Therefore, small-sized and multi-mode bent waveguide devices are highly significant and valuable for use in both photonic integrated circuits and mode division multiplexing systems. However, the design and fabrication of waveguide bends are limited by the minimum bend radius of the waveguide [20], [21], [22]. The conventional method of designing small-radius bends typically involves the use of established curve functions, such as the Euler spiral [23], [24], [25], [26] or the Bessel curve [27], [28]. Various other bend geometries, such as the hybrid bend (Euler-circle) [29], N-bend [30], Y-junction [31], corner bend [32], subwavelength grating [33], and many other methods [34], [35], [36], [37] can be used to achieve different bends. Fig. 2 illustrates some of them. However, most of these geometries use well-defined curve functions, which limits the parameter space. The shape constraint imposed by the established curve function limits the discovery of compact bends with higher performance.

In contrast, the cubic spline interpolation method is capable of defining any curves with irregular shapes. This method provides great flexibility to accommodate different bend radii and mode

Manuscript received 26 October 2023; revised 6 February 2024; accepted 24 February 2024. Date of publication 27 February 2024; date of current version 8 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 61975198, in part by the State Key Laboratory of Information Photonics and Optical Communications under Grant IPOC2021ZT11, and in part by the Fundamental Research Funds for the Central Universities under Grant XJSJ23177. (Corresponding author: Lei Zhang.)

Enge Zhang and Lei Zhang are with the State Key Laboratory of Information Photonics and Optical Communications, Beijing University of Posts and Telecommunications, Beijing 100876, China, and also with the School of Integrated Circuits, Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: zhangenge@bupt.edu.cn; zhang-lei@bupt.edu.cn).

Shanglin Yang is with the School of Optoelectronic Engineering, Xidian University, Xi'an 710126, China (e-mail: yangshanglin@xidian.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JLT.2024.3370675>.

Digital Object Identifier 10.1109/JLT.2024.3370675

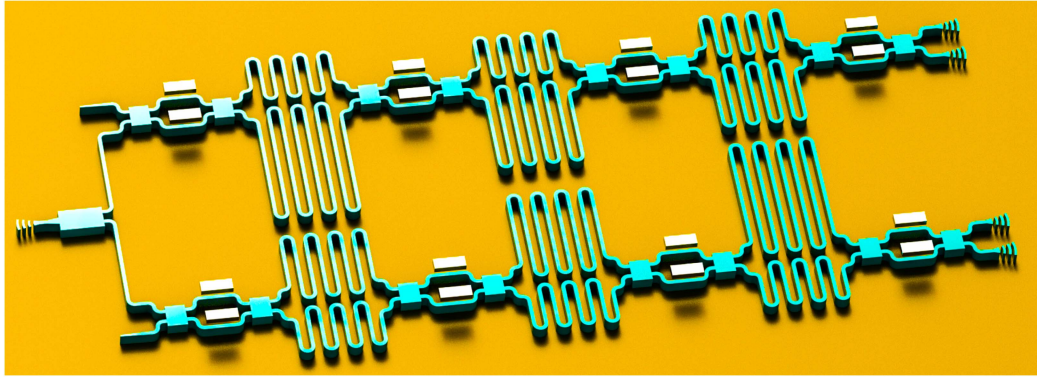


Fig. 1. Schematic of an on-chip tunable optical delay line with waveguide bends for interconnection and routing.

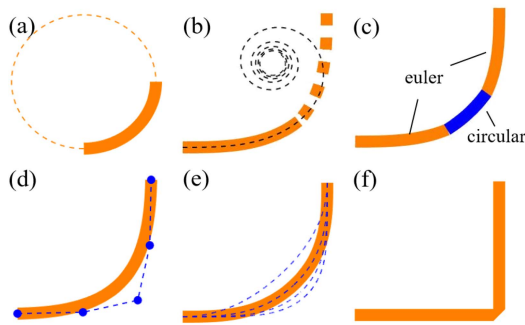


Fig. 2. Schematic of different waveguide bends reported in literature. (a) Circle bend, (b) Euler bend, (c) Euler–Circle–Euler bend, (d) Bessel bend, (e) N-bend, (f) Corner bend.

numbers and increases the likelihood of finding a bend geometry with lower insertion loss. The cubic spline interpolation method for curve fitting is based on the number of different interpolation points, which significantly affects the final shape of the curve. When selecting multiple interpolation points to design the curve shape, the traditional parameter scanning method requires considerable time and effort.

Therefore, we choose the inverse design approach to find the optimal curve shape, which is suitable for such a task with a large parameter space.

Recently, the inverse design method has become a widely used approach for developing high-performance nanophotonic devices. The inverse design method uses automated design algorithms [38], [39], [40] to alleviate the challenges of manual design, thus freeing designers from tedious work. The inverse design methodology has also been used to design waveguide bends [41], [42]. At present, inverse design algorithms mainly consist of heuristic algorithms, gradient algorithms, neural network algorithms, etc. [11], [43], [44], [45], [46], [47]. The latter two methods are typically complex and require a considerable amount of time and effort to master. In practical applications, both methods require significant parameter tuning, and the latter method also requires the acquisition of a large amount of data. In contrast, heuristic algorithms are characterized by their elegant simplicity and easy mastery. Particle swarm optimization (PSO) is a heuristic algorithm that has found many applications in various fields due to its simplicity and ease of use [48]. However,

the PSO algorithm is prone to get trapped in local optima, has low search precision, and requires considerable tuning in practical design.

To overcome the above problems, we refine the traditional PSO algorithm to balance its local and global search capabilities. After modification, it has the ability to escape local optima and converge to the global optimum even in cases where suboptimal parameters are used (at the cost of additional iterations). We refer to our improved algorithm as the self-adaptive particle swarm optimization (SAPSO) algorithm.

We have combined the cubic spline interpolation method with the SAPSO algorithm, which allows the design of low-loss bends of arbitrary width, radius and given mode. To verify the effectiveness of the method, we designed and experimentally verified single, single mode, dual mode and triple mode silicon waveguide bends with widths of 500 nm, 900 nm and 1400 nm, respectively. The effective radii of the three bends are 2.5 μm , 4 μm , and 5 μm , respectively.

The rest of this article is organized as follows. Section II describes the design principles. Section II-A explains the principle of cubic spline interpolation. Section II-B explains how to construct the waveguide bending structure using the cubic spline interpolation curves. Section II-C describes the improved particle swarm algorithm. Finally, Section II-D summarizes the steps for designing the waveguide bending using the SAPSO algorithm and the cubic spline interpolation method. Section III presents the simulation and experimental results. Lastly, Section IV provides a summary of our work.

II. DESIGN PRINCIPLE

A. Cubic Spline Interpolation

Cubic spline interpolation is a popular fitting method for creating a smooth curve that passes through a set of defined points. This curve is made up of a sequence of cubic polynomials. Compared to other interpolation polynomials, such as Lagrange polynomials and Newton polynomials, the interpolation polynomials generated by this method are smoother and have smaller errors.

Given a set of $n+1$ data points (x_i, y_i) , that follow $x_0 < x_1 < \dots < x_n$, the formula for the cubic spline interpolation function

$S(x)$ is as follows:

$$S(x) = \begin{cases} S_1(x), & x_0 \leq x < x_1 \\ S_2(x), & x_1 \leq x < x_2 \\ \vdots & \vdots \\ S_n(x), & x_{n-1} \leq x < x_n \end{cases} \quad (1)$$

For each interval between the points (x_{i-1}, y_{i-1}) and (x_i, y_i) , the cubic polynomial $S_i(x)$, $i = 1, 2, \dots, n$ is:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, \quad (2)$$

where a_i, b_i, c_i and d_i are the coefficients of the cubic polynomial $S_i(x)$. Since there are n cubic polynomials, the total number of coefficients is $4n$. When two cubic polynomial functions describe adjacent intervals, continuity at the common endpoints and equality of their first and second derivatives are required. This statement can be expressed as follows:

$$\begin{cases} S_i(x_i) = y_i \\ S_i(x_i) = S_{i+1}(x_i) \\ S'_i(x_i) = S'_{i+1}(x_i) \\ S''_i(x_i) = S''_{i+1}(x_i), \end{cases} \quad (3)$$

We derive $4n - 4$ equations from (3) for the $n - 1$ intermediate data points. We have two additional equations at the two end points (x_0, y_0) and (x_n, y_n) , i.e., $S_0(x_0) = y_0$ and $S_n(x_n) = y_n$. Thus, we have a total of $4n - 2$ equations in total. Since there are $4n$ unknown coefficients in (2), we need two more equations to calculate all the coefficients. The cubic spline interpolation uses three forms of boundary conditions that produce the two final equations. The boundary conditions include clamped, natural, and not-a-knot conditions. Here are their formulas:

$$S'_1(x_0) = P, S'_n(x_n) = Q \quad (4a)$$

$$S''_1(x_0) = 0, S''_n(x_n) = 0 \quad (4b)$$

$$S'''_1(x_0) = S'''_2(x_1), S'''_{n-1}(x_{n-1}) = S'''_n(x_n), \quad (4c)$$

where P and Q can be chosen arbitrarily. Eq (4a) through (4c) are the formulas for the clamped, natural, and not-a-knot conditions, respectively.

Selecting one of the three boundary conditions results in a total of $4n$ equations. All the coefficients can be determined from these equations, resulting in a unique determination of the spline interpolation functions.

Equation (4a) shows that the clamped boundary condition, unlike the other two boundary conditions mentioned, requires two additional parameters (P and Q) that represent the first derivatives at the endpoints. When designing a low-loss waveguide bend, our goal is to achieve a smooth transition from the straight waveguide to the curved waveguide, with minimal variation in direction, to reduce the mismatch between the two modes. Therefore, at the junction, we make sure that the first derivatives at the ends of the curved waveguides match those of the corresponding ends of the straight waveguides. To illustrate this, we have constructed the model shown in Fig. 3. In the figure, the solid line represents a straight line, while the dashed

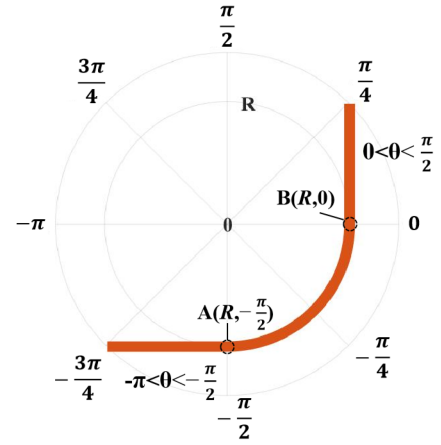


Fig. 3. Schematic diagram of interconnecting curves and lines.

line represents a circular arc. The connection points, A and B, are located at $(R, -\pi/2)$ and $(R, 0)$, respectively.

Fig. 3 shows the device structure represented in polar coordinates. The waveguides with arc degrees between $-\pi$ and $-\pi/2$ are input (output) waveguides, while those with arc degrees greater than 0 and less than $\pi/2$ are output (input) waveguides. The bending waveguides to be designed have radians between $-\pi/2$ and 0. The function of these three segments can be described by the following equation.

$$\begin{cases} r = \frac{R}{\cos(\theta + \pi/2)} & -\pi < \theta \leq -\pi/2 \\ r = R & -\pi/2 < \theta < 0 \\ r = \frac{R}{\cos(\theta)} & 0 \leq \theta < \pi/2 \end{cases} \quad (5)$$

In polar coordinates, the first derivative of the curve function at point B with respect to the polar angle θ can be expressed as follows:

$$\frac{dr}{d\theta} \Big|_{\theta=0} = \frac{R \sin(\theta)}{\cos^2(\theta)} \Big|_{\theta=0} = 0 \quad (6)$$

The same process can be done for the line connected to point A, where the first derivative is also zero. This indicates that the curves do not make sudden turns at the connection points. In the subsequent calculations that use the clamped boundary conditions in this paper, the first derivative is assigned a value of zero at both endpoints. In other words, the two parameters P and Q in (4a) are set to zero.

B. Waveguide Bend Design

Fig. 4 shows that the waveguide bend has two curves, AB and DC, which correspond to the inner and outer curves of the bend, respectively. The four solid lines represent the sidewalls of two straight waveguides connected to the waveguide bend, where the input waveguide width is the same as the output waveguide width, i.e., $W_{input} = W_{output}$. The structure of the waveguide bend can be modified by changing the trajectories of curves AB and DC. The effective radius of the waveguide bend is $R_{inner} + W_{input}/2$. The cubic spline interpolation method is used to describe the AB and DC curves. Equations (1) to (4) show that to construct curves using the cubic spline interpolation

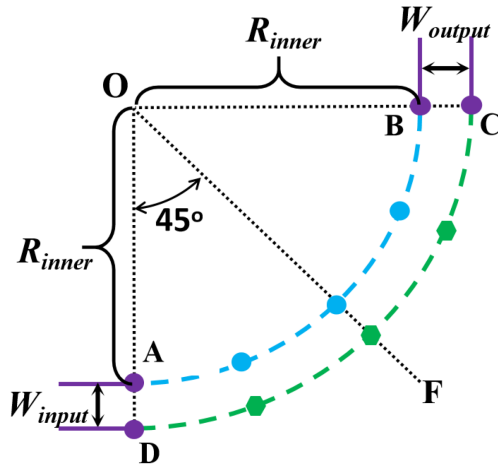


Fig. 4. Schematic diagram of the geometries of bends.

method, a group of points with established coordinates and boundary conditions is required. A sample of known points is shown in Fig. 4, which includes solid circle points placed on the curve AB. Together with the endpoints A and B, they form a group of known points located on the inner curve. In addition, the solid hexagons and points C and D define another set of known points located on the outer curve. By adjusting the positions of these points, waveguide structures with different geometries can be achieved.

To achieve a seamless connection between the curved and straight waveguides, it is necessary to keep the endpoints of the curved waveguides fixed. For example, when dealing with the inner curve, it's crucial to keep both endpoints A and B stationary while focusing only on optimizing the locations of the movable points (MPs), which are marked in blue. The same approach applies to the outer curves, where the green dots also represent MPs. In addition, the 90-degree bend has symmetry along the angle bisector OF, so that points below (or above) the line OF can be directly mirrored by those above (or below) it without any additional setup. To provide sufficient parameter space, the range of motion of the MP is chosen to be from zero to twice the target radius.

Choosing the appropriate number of MPs is a critical issue. Fig. 5 shows some examples of the fitted curves with different numbers of MPs. It can be seen that as the number of MPs increases, the geometric structure of the curve becomes more complex. Potentially, a complex curve with a larger parameter space may result in geometries that have superior performance, but also result in a larger number of less efficient structures, thus complicating our search process. In addition, the fine structures on the waveguide sidewall can be a manufacturing challenge. Therefore, in the actual design, we choose to start with a small number of MPs for the waveguide bend design, for example, one MP on each side of the inner and outer curves of the bend, and then increase the number of MPs as needed to meet our insertion loss and number of supported modes requirements. To guarantee successful production of the designed bend, the minimum size of the current structure required to meet the

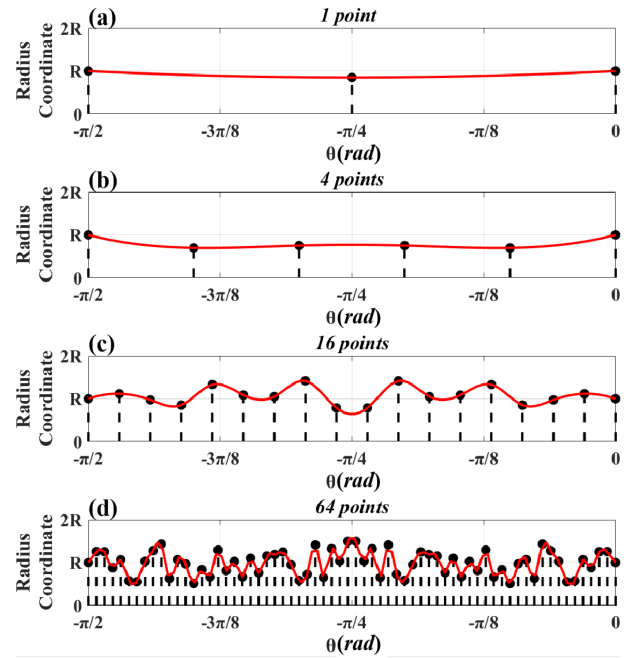


Fig. 5. Shapes of the fitted curves with (a) 1, (b) 4, (c) 16, (d) 64 moving points.

processing requirements of the factory can be determined by measuring the distance between two neighbouring MPs.

When designing the bend geometry using cubic spline interpolation, explicit boundary conditions must be specified. As shown in (4), there are three boundary conditions. The following analysis examines the effect of these conditions on the shape of the curve, which varies depending on the position of the MPs. For example, consider a single MP as shown in Fig. 6, where the left and right points $((R, -\pi/2)$ and $(R, 0)$) are fixed points. As shown in the figure, the closer the radius coordinate is to that of the fixed point, the smaller the discrepancy between the lines under different boundary conditions. Therefore, we choose to analyze the case most affected by boundary conditions, that is, those with the farthest radius coordinates (0 or $2R$).

In addition to the position of the MPs, the boundary conditions have different effects on the shape of the curve for different numbers of MPs. As shown in Fig. 7, the influence of the boundary conditions on the shape of the curve is mainly concentrated on the leftmost and rightmost segments, while the influence on the inner segments is minimal and decreases as the number of MPs increases. When the number of MPs is large, we can optimize the placement of MPs near the ends of the curved waveguide to mitigate the effects of boundary conditions. As shown in Fig. 8, by adjusting the position of the MPs, three lines with different boundary conditions are aligned.

The results in Fig. 8 show that if we have enough MPs, the choice of boundary conditions is trivial. For a curve fitted under one boundary condition, it can be reproduced under another boundary condition if we place close unequally spaced moving points on the target curve near the two fixed points. This means that with enough MPs, we can choose any of the three boundary conditions for the waveguide bend design.

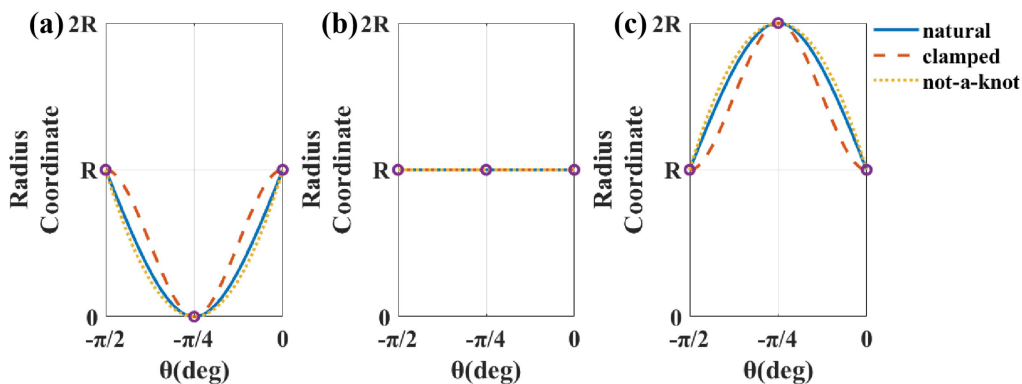


Fig. 6. Influence of boundary conditions on the shape of the curves at different positions of the moving points with polar coordinates of (a) $(0, -\pi/4)$, (b) $(R, \pi/4)$, (c) $(2R, -\pi/4)$.

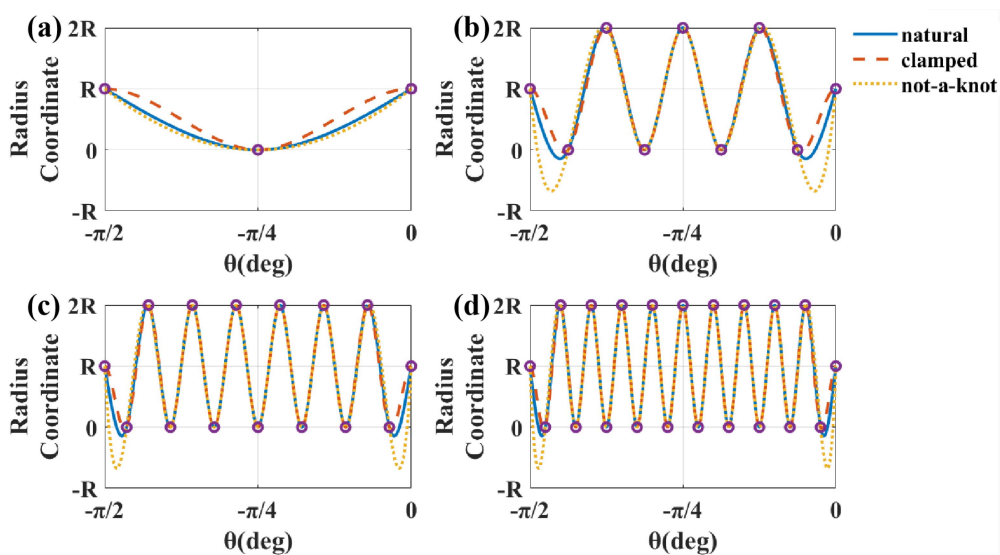


Fig. 7. Influence of boundary conditions on the shape of the curves with equally spaced moving points and moving point numbers of (a) 1, (b) 7, (c) 13, (d) 19.

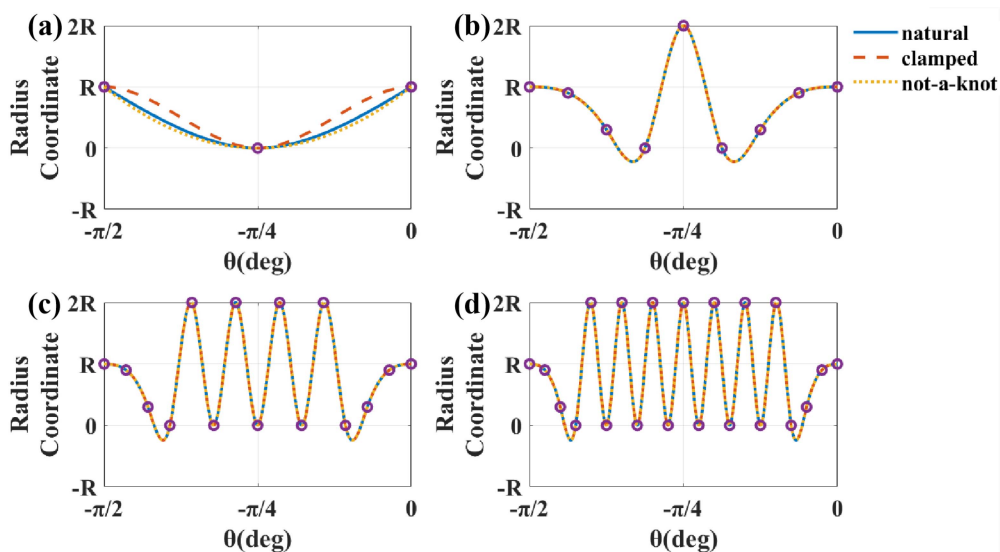


Fig. 8. Influence of boundary conditions on the shape of the curves with specially designed moving points near two fixed points and moving point numbers of (a) 1, (b) 7, (c) 13, (d) 19.

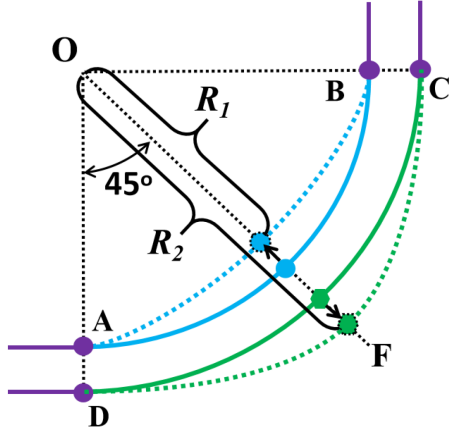


Fig. 9. Schematic diagram of MPs movement.

According to the above analysis, the effect of boundary conditions on the shape of the bend becomes more significant when the number of MPs is small. To directly observe the effect of different boundary conditions on the waveguide bend, we perform an analysis using a bend with an input and output waveguide width of 500 nm and an effective radius of 2.5 μm , while selecting one MP on each side of the bend. This waveguide bend is for single mode transmission. The results are shown in Fig. 9, where R_1 is the radius coordinate of the MP on the inside of the bend and R_2 is the radius coordinate of the MP on the outside of the bend, and the insertion loss (IL) can be calculated by the following equation:

$$IL = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \quad (7)$$

To prevent the intersection of two curves, the following conditions must be met:

$$R_2 \geq R_1 \quad (8)$$

As shown in Fig. 10, insertion loss below 0.1 dB can be achieved for single mode bending regardless of the selected boundary condition combination. Fig. 11 shows the percentage of insertion loss below 0.1 dB as R_1 and R_2 are varied for different boundary condition combinations. As shown in the figure, the probability of obtaining a device with insertion loss < 0.1 dB is greater when the clamped boundary condition is selected for the inner curve. The boundary condition on the outer curve has no significant effect in this example.

We also perform the same analysis using a bend with a width of 900 nm and an effective radius of 4 μm for dual mode transmission, while selecting one MP on each side. The results are shown in Figs. 12 and 13. As shown in Fig. 13, the probability of obtaining a device with insertion loss < 0.5 dB is greater than 3.2% when the clamped boundary condition is selected for the inner curve and the natural or clamped boundary condition is selected for the outer curve.

Based on the above analysis, in the following designs, we will first select clamped and natural boundary conditions for the inner and outer curves, respectively. If the design results do not meet the requirements, we adopt the clamped-clamped boundary

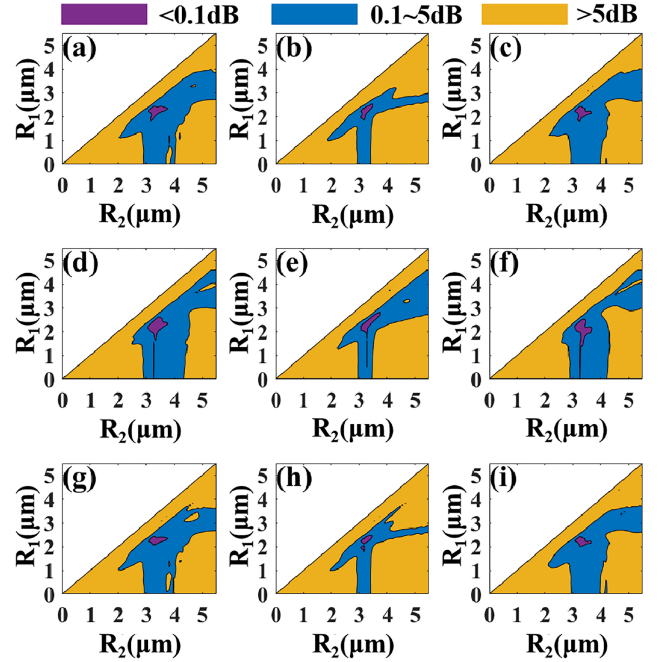


Fig. 10. Statistical analysis of single mode bending losses with different moving point positions and different boundary condition combinations (one MP for each of the inner and outer curves). The boundary conditions used for the inner & outer curves are (a) natural & natural, (b) natural & clamped, (c) natural & not-a-knot, (d) clamped & natural, (e) clamped & clamped, (f) clamped & not-a-knot, (g) not-a-knot & natural, (h) not-a-knot & clamped, and (i) not-a-knot & not-a-knot boundary conditions.

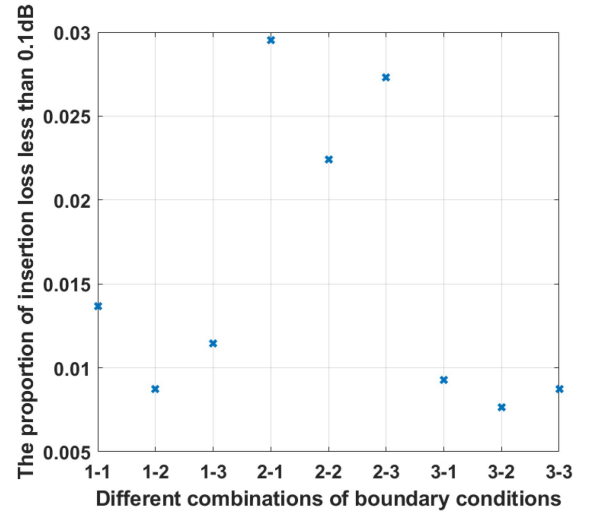


Fig. 11. Percentage of insertion loss below 0.1 dB for single mode bend as R_1 and R_2 are varied for different combinations of boundary conditions. The natural, clamped and not-a-knot boundary conditions are represented by 1, 2, 3, respectively.

condition. If the design result still fails to meet the requirements, we try other combinations.

We have analyzed the influence of MP number and boundary condition on the bend performance. The design of the waveguide bend requires given parameters such as waveguide width, effective radius, number of MPs, boundary condition, and number of

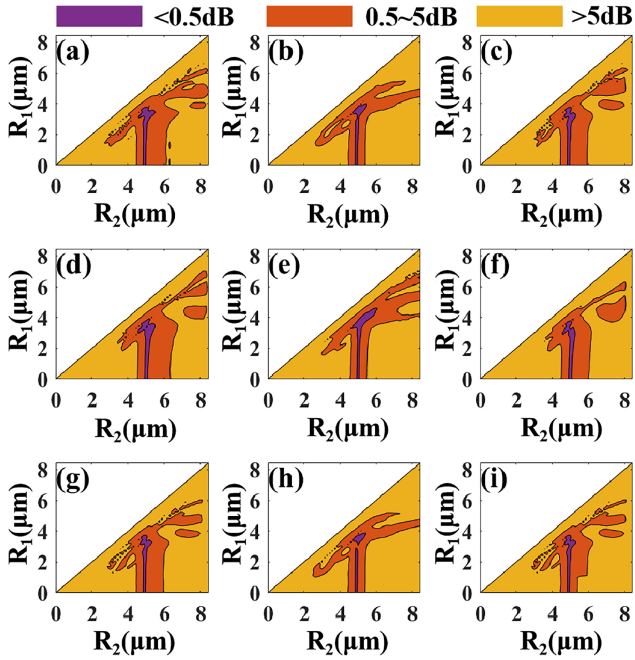


Fig. 12. Statistical analysis of dual mode bending losses with different moving point positions and different boundary condition combinations (one MP for each of the inner and outer curves). The boundary conditions used for the inner & outer curves are (a) natural & natural, (b) natural & clamped, (c) natural & not-a-knot, (d) clamped & natural, (e) clamped & clamped, (f) clamped & not-a-knot, (g) not-a-knot & natural, (h) not-a-knot & clamped, and (i) not-a-knot & not-a-knot boundary conditions.

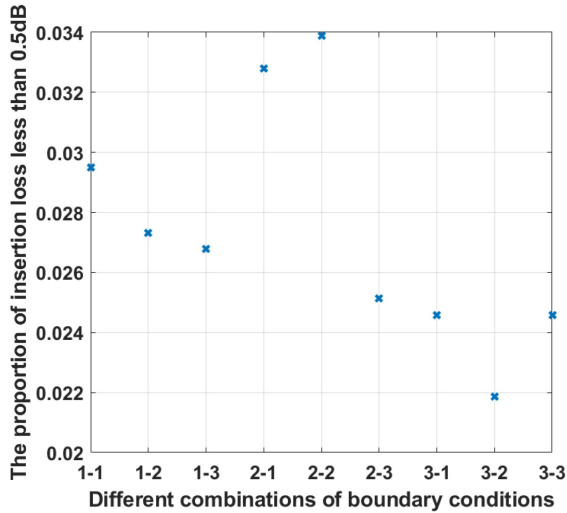


Fig. 13. Percentage of insertion loss below 0.5 dB for dual mode bend as R_1 and R_2 are varied for different combinations of boundary conditions. The natural, clamped and not-a-knot boundary conditions are represented by 1, 2, 3, respectively.

supporting modes. Finding the optimal positions for the moving points is the task of the optimization algorithm.

C. Algorithm Analysis

Particle swarm optimization (PSO) was proposed by Eberhart and Kennedy in 1995 [48]. PSO simulates the foraging behavior of birds. It maps the search space for a specific problem to the

flight space of a flock of birds. Each bird is regarded as a particle without mass and volume, which represents a possible solution to the problem. The process of finding the optimal solution of the optimization problem is just like the process of birds looking for food, which is the result of a candidate solution. PSO regards individuals in the population as particles in the N-dimensional search space. Each particle moves in the space at a certain speed and realizes the evolution of candidate solutions under the guidance of its own experience and the experience of the group. In our design, each particle represents a specific type of curved geometry. The position of each particle indicates the value of MP, while the velocity reflects the rate at which MP changes. The goal of the particle swarm optimization is to identify the curve geometry that yields the minimum insertion loss in the current configuration.

The formulas for PSO are as follows,

$$v^i(n+1) = \omega v^i(n) + c_1 r_1 (p_{best}^i - x^i(n)) + c_2 r_2 (g_{best} - x^i(n)) \quad (9)$$

$$x^i(n+1) = x^i(n) + v^i(n+1), \quad (10)$$

Where i is the current particle, n is the current number of iterations, v is the speed of particle movement, x is the current position of the particle, p_{best} is the optimal position of the individual, g_{best} is the optimal position of the group, ω is the inertia constant, c_1 is the personal best weight, c_2 is the global best weight, r_1 and r_2 are random numbers from 0 to 1.

PSO starts from the random locations given at the beginning. In the iterative process, it evaluates whether the current solution is good or not according to the figure of merit (FOM) of the problem to be solved. Finally, it iteratively finds the global optimal solution. In each iteration, the optimal value and corresponding position found in the history of a single particle will be recorded as the individual experience of the particle. The optimal values and corresponding positions found in the history of all particles will also be recorded as the experience of the population. The direction of motion of each particle will be determined by individual experience and population experience. Among them, the proportion of individual experience and population experience in each movement speed will change randomly due to the existence of random numbers r_1 and r_2 , thus bringing certain randomness to the algorithm.

In order to ensure a comprehensive search of the parameter space, the algorithm must possess global search capabilities to locate the range where optimal values reside, which necessitates greater speed. However, once the optimal value range has been determined, the algorithm must employ local search abilities to pinpoint its exact location, requiring slightly less speed. Regarding the particle search process, the algorithm's early stage should exhibit strong global search capabilities but weak local search abilities. Conversely, in the algorithm's later stages, its local search abilities are robust, while its global search capabilities weaken. However, in traditional PSO, the global and local search abilities are fixed once the algorithm parameters are set. Therefore, achieving good FOM requires extensive parameter tuning.

In addition to the aforementioned issue, (9) reveals that as the number of iterations increases, the individual experience gradually approaches group experience, leading to a decline in velocity. If the swarm consistently fails to locate a new population optimal position, which means the group has fallen into a local optimum, velocity will approach zero and the algorithm will converge on its current population optimal position. Hence, it is indeterminate whether the ultimate outcome attains local optimality or global optimality. We have introduced a simple modification to the traditional PSO algorithm and proposed a self-adapted particle swarm optimization (SAPSO) algorithm, which address the aforementioned issues. The revised formula is presented below.

$$v^i(n+1) = \omega v^i(n) + c_1 r_1 (p_{best}^i - x^i(n)) + c_2 r_2 (g_{best} - x^i(n)) + G \mathbf{r} \quad (11)$$

$$x^i(n+1) = \alpha g_{best} + (1 - \alpha) x^i(n) + v^i(n+1) \quad (12)$$

$$\omega = \begin{cases} \omega_s + (\omega_e - \omega_s) \frac{f_g}{f_{aim}} & f_g \leq f_{aim} \\ 0 & f_g > f_{aim} \end{cases} \quad (13)$$

Where G is the random factor, α is the global best weight, $\mathbf{r}$ is a random vector in which each dimension is a random number from -1 to 1 , f_g is the group optimum, f_{aim} is the value we set close to our target FOM, ω_s is the start of inertia weight, ω_e is the end of inertia weight, which satisfies $f_g = f_{aim}$. The random factor G and global best weight α are defined as follows:

$$G = G_s + (G_e - G_s) \frac{f_g}{f_{target}} \quad (14)$$

$$\alpha = \frac{i - 1}{N - 1} \quad (15)$$

where f_{target} is our target FOM, G_s is the start of inertia weight, G_e is the inertia weight designated for the iteration, which satisfies $f_g = f_{target}$ and N is the total number of particles in the population. By comparing (9) with (11), it can be observed that the latter contains an additional stochastic term, which prevents the velocity from approaching zero and thus endows the particle to escape local optima. By comparing (10) and (12), it can be observed that the starting position of each particle in every iteration is not merely the end position of its previous iteration, but rather a combination of the group's optimal position and its own last iteration's position. This point also violates the traditional PSO algorithm's simulation of real-world behaviour. In a computer simulation, particles do not need to be in continuous motion, and the starting position of the next moment does not need to be the ending position of the previous moment. Instead, particles can start from a position that is more likely to yield a good result. To prevent the algorithm from falling into a local optimum, each particle should have a different α , as shown in (15). Additionally, this approach prevents the algorithm from getting stuck in a local optimum as a result of group experience dominance. This model improves the algorithm's efficiency, particularly for particles with sub-optimal results in the previous iteration, by relying on both group and individual experience. This prevents them from converging to a specific position. Concerning about the potential

deterioration of global search capability resulting from group experience is unnecessary. Firstly, the iteration process involves different particles with varying starting positions, thereby expanding the search scope. Secondly, (11) incorporates a random factor G that further enhances the algorithm's ability to conduct global searches. Equations (13) and (14) ensure that the algorithm possesses automatic adjustment capabilities. Parameters will be automatically adjusted based on the current calculation progress to optimize search capabilities for different stages of computation. The algorithm thus requires minimal parameter tuning, resulting in significant time and effort savings.

D. Design Methodology

The design of curve shape using cubic spline interpolation requires two prerequisites: explicit boundary conditions and MPs. The analysis in Section II-B indicates that the clamped boundary condition is employed for curves located on the inner side of the bend, while the not-a-knot boundary condition is utilized for curves situated on the outer side of the bend. If it is determined that the design requirements cannot be fulfilled, adjustments will be made in accordance with the value size relationship depicted in Fig. 5. Based on our design expertise, it is generally unnecessary to adjust the boundary conditions. According to the analysis in Section II-C, the known points consist of MPs and endpoints, with the algorithm optimizing the coordinates of MPs. To simplify algorithm design, polar coordinates are adopted for representing MP coordinates while only modifying their radius coordinates by fixing their angle coordinates. Initially, a small number of MPs is selected and increased if necessary to meet design requirements. To summarize, the design process for a waveguide bend is as follows:

- 1) Determine the width, radius, and number of modes of the bend to be designed.
- 2) Set an initial number of MPs, such as one MP each on the inside and outside of the bend, and set the clamped and not-a-knot boundary conditions inside and outside the bend respectively.
- 3) Establish the model structure and the MPs' coordinates automatically through SAPSO.
- 4) If the design outcome fails to meet the requirements, increase the number of MPs and repeat step (3). If the requirements are still not satisfied, adjust the boundary conditions as indicated in the aforementioned analysis.

It is worth noting that the definition of FOM employs a simplistic approach whereby the average insertion loss of all target modes is calculated.

III. SIMULATION AND EXPERIMENTAL RESULTS

We use the above method to design three small-radius 90-degree bends: a radius of $2.5 \mu\text{m}$ and a width of 500 nm , which supports only TE_0 mode; a radius of $4 \mu\text{m}$ and a width of 900 nm , which supports both TE_0 and TE_1 modes; and a radius of $5 \mu\text{m}$ and a width of $1.4 \mu\text{m}$, which accommodates TE_0 , TE_1 , and TE_2 modes. We use the 3D finite-difference time-domain (FDTD) method for simulation design. The results of our optimization process using SAPSO and PSO algorithms

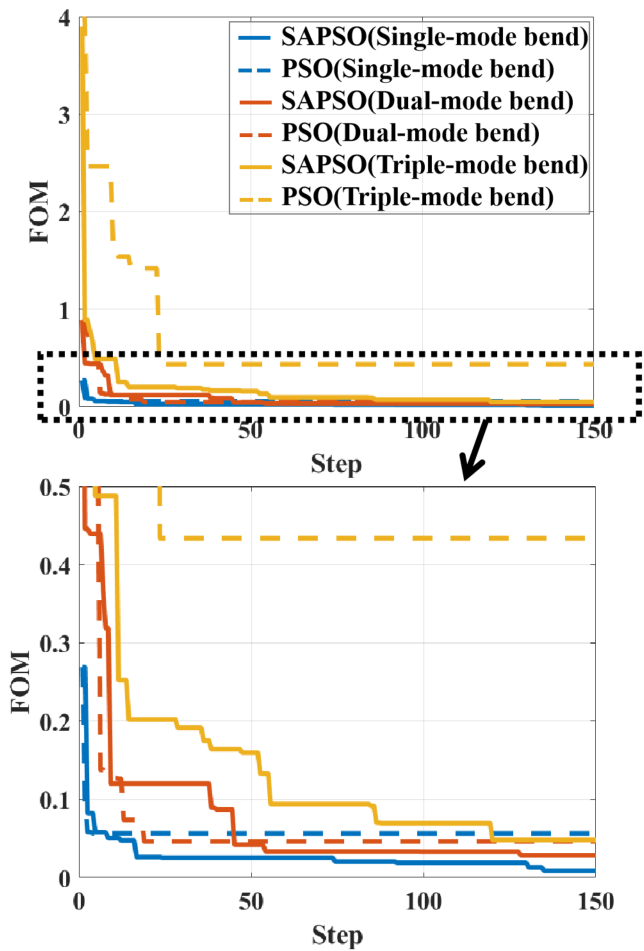


Fig. 14. Results of the optimization process using SAPSO (top) and PSO (bottom) algorithms for the three bends.

for the three bends are shown in Fig. 14. The figure clearly shows the superior performance of the SAPSO algorithm compared to the traditional PSO algorithm in terms of the final insertion loss. In addition, during the design process of three waveguide bends, the SAPSO algorithm continues to improve iteratively, while the traditional PSO algorithm converges to a local optimum within 50 iterations.

We have also employed circular, Euler, Bessel and hybrid curves to design waveguide bends with the same width, equivalent radius and number of modes. We compared the outcomes produced by our cubic spline interpolation approach to those obtained using the conventional method. Figs. 15, 16, and 17 display the comparison results for waveguide bend with single-mode, dual-mode, and triple-mode, correspondingly. Based on the figures, it is apparent that the

waveguide bends designed utilizing the cubic spline interpolation method exhibit the smallest loss and lowest crosstalk among all three cases. The figure portrays outcomes for the greatest insertion loss and the highest crosstalk. During the simulation stage, 11 wavelength points in the C-band are chosen for the simulation. Note that the markers on the lines in the figure serve solely to distinguish the different curves.

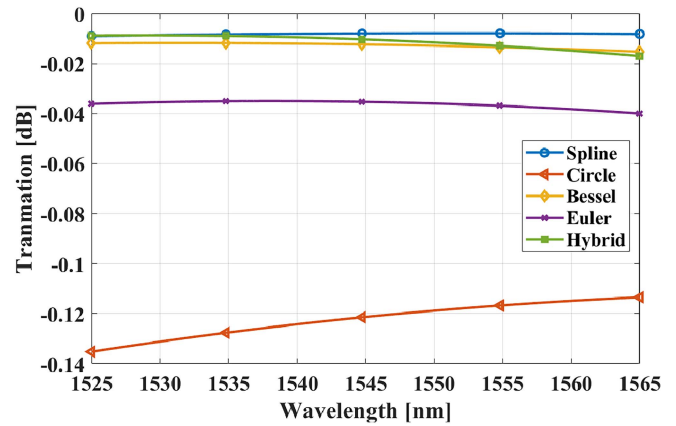


Fig. 15. Comparison results for waveguide bend with single-mode using cubic spline interpolation, circular, Euler, Bessel and hybrid curves.

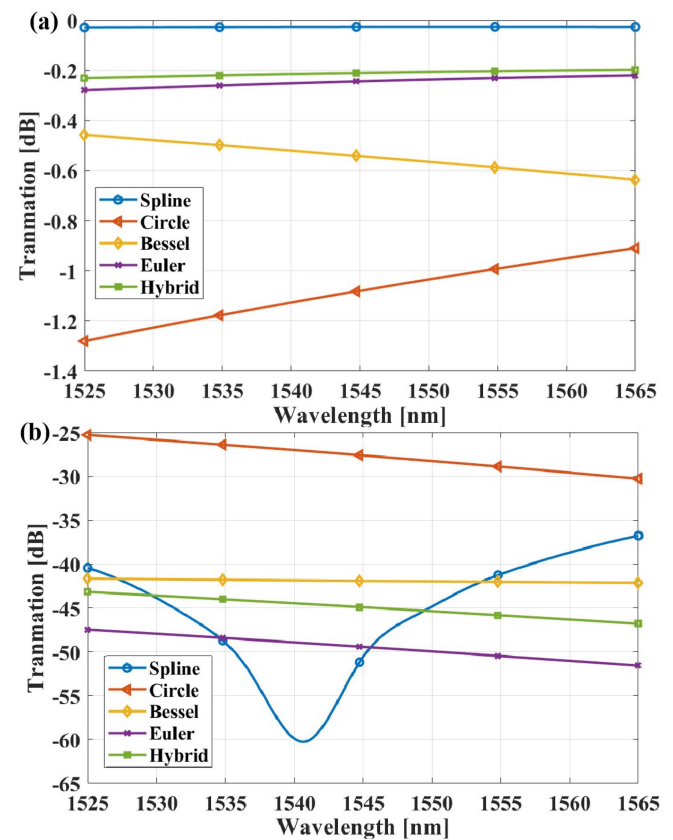


Fig. 16. Comparison results for waveguide bend with dual-mode using cubic spline interpolation, circular, Euler, Bessel and hybrid curves. (a) Maximum transmission loss, (b) Maximum crosstalk.

The devices are fabricated by e-beam lithography and dry etching. Figs. 18, 19 and 20 show the simulated and experimental results of the three bends, where the number of MPs used in the design process is 1, 1 and 10, respectively. The measured (simulated) insertion losses of the single-mode bend, the dual-mode bend and the triple-mode bend are less than 0.012 (0.009) dB, 0.04 (0.028) dB and 0.059 (0.048) dB per 9° , respectively. The measured (simulated) crosstalk between modes of the last two bends is less than -23.6 (-36.1) dB and

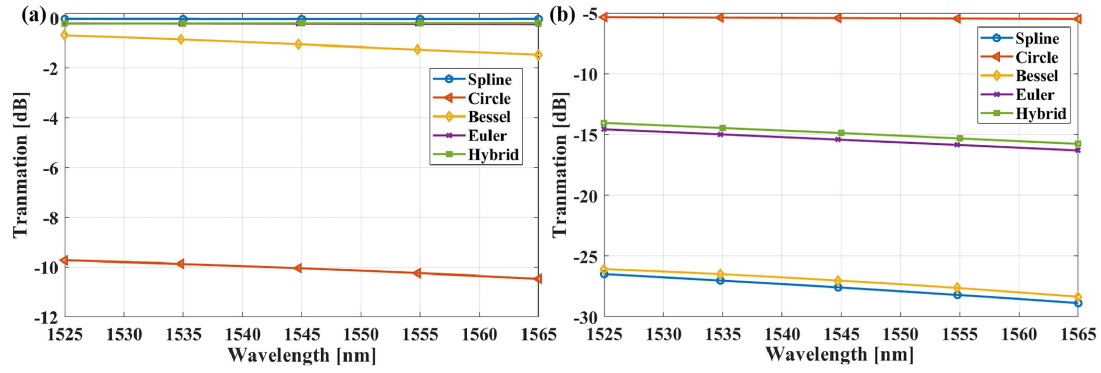


Fig. 17. Comparison results for waveguide bend with triple-mode using cubic spline interpolation, circular, Euler, Bessel and hybrid curves. (a) Maximum transmission loss, (b) Maximum crosstalk.

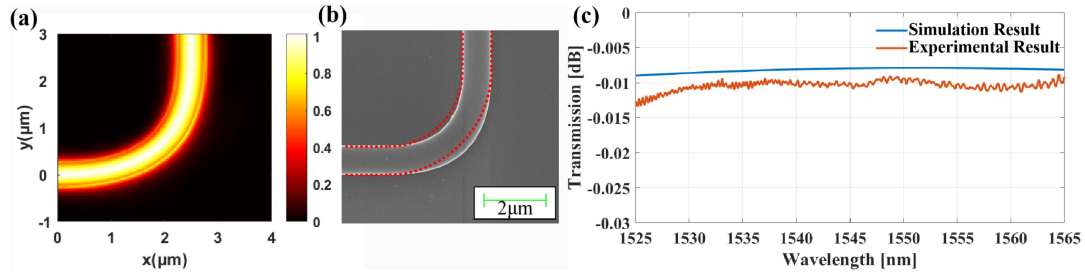


Fig. 18. Simulated and experimental results for a waveguide width of 500 nm and an equivalent bending radius of 2.5 μm . (a) Simulated electric field distribution. (b) The SEM image of the waveguide bend with the dotted line indicating a circular arc. (c) Comparison between simulated and experimental results.

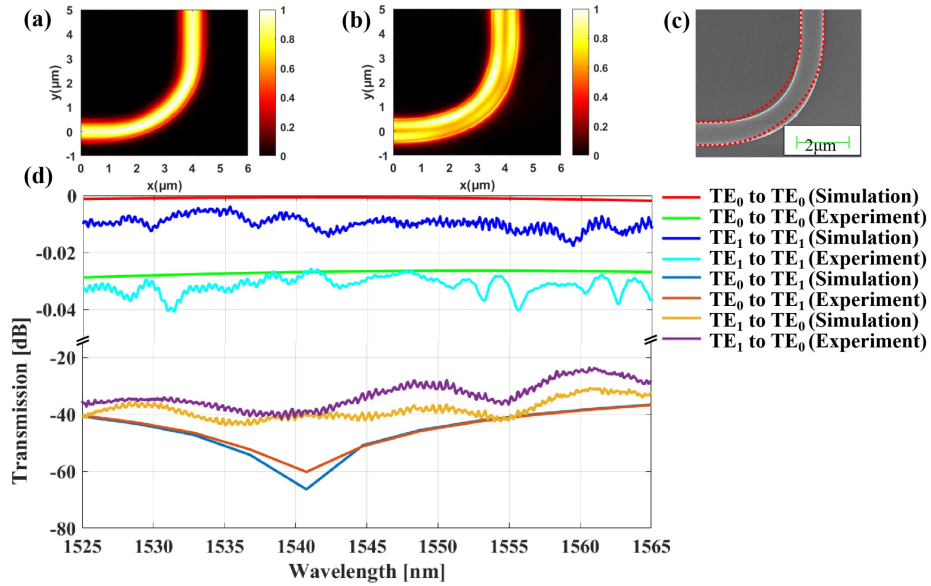


Fig. 19. Simulated and measured results were obtained for a waveguide width of 900 nm and an equivalent bending radius of 4 μm . Simulated electric fields of (a) the TE_0 mode and (b) the TE_1 mode. (c) The SEM image of the waveguide bend with the dotted line indicating a circular arc, (d) Simulated and experimental results.

−19.7 (−26.5) dB per 9° , respectively. Among the three bends, the shapes of the single-mode and dual-mode bends resemble circular arcs. We have shown the shape of an arc with a red dotted line. It is clear that our waveguide is different from the shape of a circular arc.

As can be seen from Figs. 18–20, the results of simulation and experiment are quite consistent, and the designed three bends

can maintain exceptional performance throughout the C-band, demonstrating wavelength insensitivity characteristics. We also compared our results with those of other researchers, as shown in Table I. As shown in the table, our device exhibits lower insertion loss when considering waveguide bends with the same number of modes and similar radius. In addition, our proposed design method shows broader applicability and can be used to design

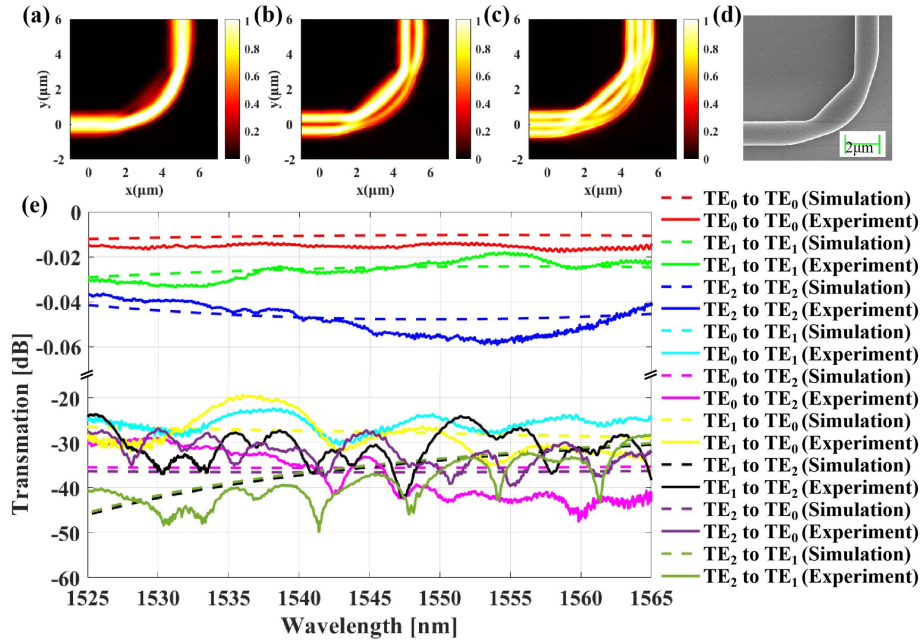


Fig. 20. Simulated and measured results were obtained for a waveguide width of 900 nm and an equivalent bend radius of 5 μm . Simulated electric fields of (a) the TE_0 mode, (b) the TE_1 mode and (c) the TE_2 mode (d) The SEM image of the waveguide bend. (e) Simulated and experimental results.

TABLE I
COMPARISON OF COMPACT WAVEGUIDE BENDS

Ref.	Radius (μm)	Number of modes	Insertion loss	Crosstalk (dB)
[30]	1	1	0.0689	—
	2	1	0.0167	—
[32]	7	2	0.18	−36
[35]	9.35	3	0.04	−29
[36]	4.4	3	0.14	−19.1
[37]	17	4	0.1	−20
	2.5	1	0.012	—
this work	4	2	0.04	−23.6
	5	3	0.059	−19.7

waveguide bends with arbitrary radius, width, and number of supported modes using the same design flow.

IV. CONCLUSION

In conclusion, we propose an efficient approach to waveguide curve design that uses cubic spline interpolation and an improved particle swarm optimization algorithm to determine the curve geometry. The proposed method not only provides a larger parameter space compared to traditional design methods, but also exhibits remarkable flexibility in designing waveguides with arbitrary widths, radii, and mode numbers (supported by the waveguide) according to the specific requirements. Meanwhile, the SAPSO algorithm based on the traditional PSO algorithm proposed in this paper avoids the problem of falling into local extremums, and can still find the global optimal value when the algorithm parameters are not appropriate (at the cost of requiring more iterations). This feature makes the design process very simple compared to other inverse design methods. At the same time, due to the simplicity of the structure, the algorithm can also

reach the optimal result quickly. Using the design method, we have successfully designed single-mode, dual-mode, and triple-mode waveguide bends with radii of 2.5 μm , 4 μm , and 5 μm , respectively. The device was fabricated by e-beam lithography, which gives excellent experimental results and is in agreement with simulations.

The proposed method is simple, efficient, and versatile in accommodating different widths, radii, and modes for waveguide bending. Its potential for large-scale photon integration makes it a significant contribution to the advancement of this field.

IV. ACKNOWLEDGMENT

The authors acknowledge the use of the AE8600 optical spectrum analyzer, and associated support services provided by Tianjin Deviser Electronics Instrument Company, Ltd.

REFERENCES

- [1] S. Rumley, D. Nikolova, R. Hendry, Q. Li, D. Calhoun, and K. Bergman, "Silicon photonics for exascale systems," *J. Lightw. Technol.*, vol. 33, no. 3, pp. 547–562, Feb. 2015, doi: [10.1109/JLT.2014.2363947](https://doi.org/10.1109/JLT.2014.2363947).
- [2] C. R. Doerr, "Silicon photonic integration in telecommunications," *Front. Phys.*, vol. 3, pp. 137–179, 2015, doi: [10.3389/fphy.2015.00037](https://doi.org/10.3389/fphy.2015.00037).
- [3] J. W. Silverstone, D. Bonneau, J. L. O'Brien, and M. G. Thompson, "Silicon quantum photonics," *IEEE J. Sel. Topics Quantum Electron.*, vol. 22, no. 6, pp. 390–402, Nov./Dec. 2016, doi: [10.1109/JSTQE.2016.2573218](https://doi.org/10.1109/JSTQE.2016.2573218).
- [4] K. de Vos, I. Bartolozzi, E. Schacht, P. Bienstman, and R. Baets, "Silicon-on-insulator microring resonator for sensitive and label-free biosensing," *Opt. Exp.*, vol. 15, no. 12, pp. 7610–7615, 2007, doi: [10.1364/OE.15.007610](https://doi.org/10.1364/OE.15.007610).
- [5] D. Richardson, J. Fini, and L. Nelson, "Space-division multiplexing in optical fibres," *Nature Photon.*, vol. 7, no. 5, pp. 354–362, 2013, doi: [10.1038/nphoton.2013.94](https://doi.org/10.1038/nphoton.2013.94).
- [6] C. Li, D. Liu, and D. Dai, "Multimode silicon photonics," *Nanophotonics*, vol. 8, no. 2, pp. 227–247, 2019, doi: [10.1515/nanoph-2018-0161](https://doi.org/10.1515/nanoph-2018-0161).
- [7] G. Li, N. Bai, N. Zhao, and C. Xia, "Space-division multiplexing: The next frontier in optical communication," *Adv. Opt. Photon.*, vol. 6, no. 4, pp. 413–487, 2014, doi: [10.1364/AOP.6.000413](https://doi.org/10.1364/AOP.6.000413).

- [8] G. Chen, Y. Yu, and X. Zhang, "Monolithically mode division multiplexing photonic integrated circuit for large-capacity optical interconnection," *Opt. Lett.*, vol. 41, no. 15, pp. 3543–3546, 2016, doi: [10.1364/OL.41.003543](#).
- [9] H. Jia et al., "WDM-compatible multimode optical switching system-on-chip," *Nanophotonics*, vol. 8, no. 5, pp. 889–898, 2019, doi: [10.1515/nanoph-2019-0005](#).
- [10] C. L. Sun, Y. Yu, G. Y. Chen, and X. L. Zhang, "Silicon mode multiplexer processing dual-path mode-division multiplexing signals," *Opt. Lett.*, vol. 41, no. 23, pp. 5511–5514, Dec. 2016, doi: [10.1364/OL.41.005511](#).
- [11] W. Chang et al., "Ultra-compact mode (de) multiplexer based on subwavelength asymmetric y-junction," *Opt. Exp.*, vol. 26, no. 7, pp. 8162–8170, Apr. 2018, doi: [10.1364/OE.26.008162](#).
- [12] D. Dai et al., "10-channel mode (de)multiplexer with dual polarizations," in *Laser Photon. Rev.*, vol. 12, no. 1, 2018, Art. no. 1700109, doi: [10.1002/lpor.201700109](#).
- [13] S. Y. Li et al., "Universal multimode waveguide crossing based on transformation optics," *Optica*, vol. 5, no. 12, pp. 1549–1556, Dec. 2018, doi: [10.1364/OPTICA.5.001549](#).
- [14] H. Xu and Y. Shi, "Metamaterial-based Maxwell's fisheye lens for multimode waveguide crossing," *Laser Photon. Rev.*, vol. 12, no. 10, 2018, Art. no. 1800094, doi: [10.1002/lpor.201800094](#).
- [15] W. Chang et al., "Ultracompact dual-mode waveguide crossing based on subwavelength multimode-interference couplers," *Photon. Res.*, vol. 6, no. 7, pp. 660–665, Jul. 2018, doi: [10.1364/PRJ.6.000660](#).
- [16] L. Han, B. P. Kuo, N. Alic, and S. Radic, "Ultra-broadband multimode 3 dB optical power splitter using an adiabatic coupler and a y-branch," *Opt. Exp.*, vol. 26, no. 11, pp. 14800–14809, May 2018, doi: [10.1364/OE.26.014800](#).
- [17] H. Xu and Y. Shi, "Ultra-broadband dual-mode 3 dB power splitter based on a y-junction assisted with mode converters," *Opt. Lett.*, vol. 41, no. 21, pp. 5047–5050, Nov. 2016, doi: [10.1364/OL.41.005047](#).
- [18] B. Wohlfeil, G. Rademacher, C. Stamatiadis, K. Voigt, L. Zimmermann, and K. Petermann, "A two-dimensional fiber grating coupler on SOI for mode division multiplexing," *IEEE Photon. Technol. Lett.*, vol. 28, no. 11, pp. 1241–1244, Jun. 2016, doi: [10.1109/LPT.2016.2514712](#).
- [19] Y. X. Lai, Y. Yu, S. N. Fu, J. Xu, P. P. Shum, and X. L. Zhang, "Compact double-part grating coupler for higher-order mode coupling," *Opt. Lett.*, vol. 43, no. 13, pp. 3172–3175, Jul. 2018, doi: [10.1364/OL.43.003172](#).
- [20] D. Dai and J. E. Bowers, "Silicon-based on-chip multiplexing technologies and devices for peta-bit optical interconnects," *Nanophotonics*, vol. 3, no. 4/5, pp. 283–311, 2014, doi: [10.1515/nanoph-2013-0021](#).
- [21] A. J. Zilkie et al., "Multi-micron silicon photonics platform for highly manufacturable and versatile photonic integrated circuits," in *IEEE J. Sel. Topics Quantum Electron.*, vol. 25, no. 5, Sep./Oct. 2019, Art. no. 8200713, doi: [10.1109/JSTQE.2019.2911432](#).
- [22] T. Aalto et al., "Open-access 3- μm SOI waveguide platform for dense photonic integrated circuits," in *IEEE J. Sel. Topics Quantum Electron.*, vol. 25, no. 5, 2019, Art. no. 8201109, doi: [10.1109/JSTQE.2019.2908551](#).
- [23] M. Cherchi, S. Ylinen, M. Harjanne, M. Kapulainen, and T. Aalto, "Dramatic size reduction of waveguide bends on a micron-scale silicon photonic platform," *Opt. Exp.*, vol. 21, no. 15, pp. 17814–17823, 2013, doi: [10.1364/OE.21.017814](#).
- [24] T. Aalto, M. Harjanne, M. Cherchi, and S. Ylinen, "Euler bends and TIR mirrors for ultra-dense PIC integration on SOI," in *Proc. 18th Eur. Conf. Integr. Opt.*, 2016. [Online]. Available: <http://www.ecio-conference.org/wp-content/uploads/2016/06/ECIO-o-20.pdf>
- [25] X. Ji et al., "Compact, spatial-mode-interaction-free, ultralow-loss, nonlinear photonic integrated circuits," in *Proc. Eur. Conf. Opt. Commun.*, Basel, Switzerland, 2022, pp. 1–4.
- [26] F. Gao et al., "Low-loss and compact bends on multi-layer SiN-on-SOI platform for photonic integrated circuits," *IEEE Photon. Technol. Lett.*, vol. 33, no. 20, pp. 1131–1134, Oct. 2021, doi: [10.1109/LPT.2021.3107865](#).
- [27] H. P. Bazargani, J. Flueckiger, L. Chrostowski, and J. Azaña, "Microring resonator design with improved quality factors using quarter Bezier curves," in *Proc. Conf. Lasers Electro-Optics*, San Jose, CA, USA, 2015, pp. 1–2, doi: [10.1364/CLEO_AT.2015.JTu5A.58](#).
- [28] C. L. Manganelli et al., "Design of coupled micro-ring resonators for silicon photonic switching matrices," in *Proc. IEEE Opt. Interconnects Conf.*, San Diego, CA, USA, 2016, pp. 84–85, doi: [10.1109/OIC.2016.7483008](#).
- [29] T. Fujisawa, S. Makino, T. Sato, and K. Saitoh, "Low-loss compact and fabrication-tolerant Si-wire 90° waveguide bend using clothoid and normal curves for large scale photonic integrated circuits," *Opt. Exp.*, vol. 25, no. 8, pp. 9150–9159, 2017, doi: [10.1364/OE.25.009150](#).
- [30] L. Zhang et al., "Ultra-compact low-loss modified n-adjustable bends," *J. Lightw. Technol.*, vol. 41, no. 9, pp. 2792–2800, May 2023, doi: [10.1109/JLT.2023.3237571](#).
- [31] Y. A. Vlasov and S. J. McNab, "Losses in single-mode silicon-on-insulator strip waveguides and bends," *Opt. Exp.*, vol. 12, no. 8, pp. 1622–1631, 2004, doi: [10.1364/OPEX.12.001622](#).
- [32] Y. Wang and D. Dai, "Multimode silicon photonic waveguide corner-bend," *Opt. Exp.*, vol. 28, no. 7, pp. 9062–9071, 2020, doi: [10.1364/OE.387978](#).
- [33] H. Wu et al., "Ultra-sharp multimode waveguide bends with sub-wavelength gratings," in *Laser Photon. Rev.*, vol. 13, no. 2, 2019, Art. no. 1800119, doi: [10.1002/lpor.201800119](#).
- [34] S. Sun et al., "Ultra-sharp silicon multimode waveguide bends based on double free-form curves," *Photon. Res.*, vol. 10, no. 6, pp. 1484–1490, 2022, doi: [10.1364/PRJ.445529](#).
- [35] S. Sun et al., "Inverse design of ultra-compact multimode waveguide bends based on the free-form curves," in *Laser Photon. Rev.*, vol. 15, no. 9, 2021, Art. no. 2100162, doi: [10.1002/lpor.202100162](#).
- [36] S. Yang et al., "CMOS-compatible ultra-compact silicon multimode waveguide bend based on inverse design method," in *Opt. Commun.*, vol. 523, 2022, Art. no. 128733, doi: [10.1016/j.optcom.2022.128733](#).
- [37] S. Li et al., "Compact and broadband multimode waveguide bend by shape-optimizing with transformation optics," *Photon. Res.*, vol. 8, no. 12, pp. 1843–1849, 2020, doi: [10.1364/PRJ.403818](#).
- [38] A. Y. Piggott et al., "Inverse design and demonstration of a compact and broadband on-chip wavelength demultiplexer," *Nature Photon.*, vol. 9, no. 6, pp. 374–377, 2015, doi: [10.1038/nphoton.2015.69](#).
- [39] Z. Liu et al., "Integrated nanophotonic wavelength router based on an intelligent algorithm," *Optica*, vol. 6, no. 10, pp. 1367–1373, 2019, doi: [10.1364/OPTICA.6.001367](#).
- [40] S. Yang et al., "Gradient-probability-driven discrete search algorithm for on-chip photonics inverse design," *Opt. Exp.*, vol. 29, no. 18, pp. 28751–28766, 2021, doi: [10.1364/OE.432313](#).
- [41] Z. Zhang et al., "Design of compact, broadband, and low-loss silicon waveguide bends with radius under 500 nm," in *Photonics*, vol. 9, no. 9, 2022, Art. no. 616, doi: [10.3390/photonics9090616](#).
- [42] C. Sun, Y. Yu, G. Chen, and X. Zhang, "Ultra-compact bent multimode silicon waveguide with ultralow inter-mode crosstalk," *Opt. Lett.*, vol. 42, no. 15, pp. 3004–3007, 2017, doi: [10.1364/OL.42.003004](#).
- [43] D. Liu, Y. Tan, E. Khoram, and Z. Yu, "Training deep neural networks for the inverse design of nanophotonic structures," *ACS Photon.*, vol. 5, no. 4, pp. 1365–1369, 2018, doi: [10.1021/acsp Photonics.7b01377](#).
- [44] K. Wang et al., "Inverse design of digital nanophotonic devices using the adjoint method," *Photon. Res.*, vol. 8, no. 4, pp. 528–533, 2020, doi: [10.1364/PRJ.383887](#).
- [45] B. Shen, P. Wang, R. Polson, and R. Menon, "An integrated-nanophotonics polarization beamsplitter with $2.4 \times 2.4 \mu\text{m}^2$ footprint," *Nature Photon.*, vol. 9, no. 6, pp. 378–382, 2015, doi: [10.1038/nphoton.2015.80](#).
- [46] J. Huang et al., "Implementation of on-chip multi-channel focusing wavelength demultiplexer with regularized digital metamaterials," *Nanophotonics*, vol. 9, no. 1, pp. 159–166, 2020, doi: [10.1515/nanoph-2019-0368](#).
- [47] S. Molesky et al., "Inverse design in nanophotonics," *Nature Photon.*, vol. 12, no. 11, pp. 659–670, 2018, doi: [10.1038/s41566-018-0246-9](#).
- [48] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. Int. Conf. Neural Netw.*, Perth, WA, Australia, 1995, pp. 1942–1948, doi: [10.1109/ICNN.1995.488968](#).